

Nguyen Quoc Duy

0979.783.396 | quocduy0322@gmail.com | www.duyaivy.id.vn

OVERVIEW

Software Engineering student with hands-on experience through internship and team projects. Experienced in developing, debugging, testing, and deploying software in collaborative environments. Interested in problem solving, continuous learning, and adapting to new technologies, with a focus on building reliable software and using AI tools effectively to support the development process.

EDUCATION

Da Nang University of Science and Technology (DUT)

Information Technology – Major: Software Engineering
GPA: 3.83 / 4.0

Sep 2023 – Present

Anticipated graduation: 04/2027

EXPERIENCE

Intern Frontend Developer

Nov 2025 – Feb 2026

AIAIVN - Vietnam Artificial Intelligence Joint Stock Company

4 months

- Contributed to **HeriStepAI GPS** by fixing bugs and implementing simple features for real-world deployments in Hoi An and the Da Nang New Year Marathon 2025–2026.
- Learned and worked with the existing **HeriStepAI Beacon** and supported the project deployment at the Da Nang Museum of Cham Sculpture.
- Built **Website Payment**, a web application that supports the ticket purchasing flow by redirecting users from the HeriStepAI GPS platform to complete payment.
- Worked within an existing software development workflow involving task coordination, bug fixing, feature development, testing, deployment support, and team collaboration.

Frontend Developer — Student Community

Sep 2024 – Sep 2026

GDGoC DUT– Google Developer Groups On Campus DUT

2 years

- Shared technical knowledge and supported peer learning within the student technology community.
- Developed interactive mini games such as Lucky Wheel, Memory Matching Game,.. to increase engagement during club events and activities.
- Collaborated with club members on technology projects for learning, community-oriented initiatives, and participation in hackathons and technology competitions.

PROJECTS

Smart Calorie Estimation & Meal Recommendation Platform

Dec 2025 – Apr 2026

| React Native, Expo, Express.js, PostgreSQL, Redis, Firebase, MQTT

5 months

- Collaborated in a 4-member team to develop a full-stack nutrition platform combining food recognition, weight-based calorie estimation, and personalized meal recommendations.
- Developed mobile features, integrating backend APIs, state management, and push notifications.
- Contributed to backend APIs, authentication and MQTT-based communication with an IoT smart scale.
- Integrated AI-based food recognition with smart-scale weight data to support calorie estimation.
- [Github Backend](#) – [Github Mobile](#) – [Demo](#)

Restaurant Management Platform

Mar 2026 – Jun 2026

| Next.js, Django REST Framework, Socket.IO, Redis, OpenAI

3 months

- Collaborated in a 2-member team to develop a full-stack restaurant platform covering the end-to-end ordering flow, from menu browsing and checkout to VNPay payment and order tracking.
- Implemented real-time order and service status updates using Socket.IO and Redis, synchronizing changes between customer and restaurant management interfaces.
- Integrated an OpenAI-powered chatbot for customer assistance and dish recommendations, with Vietnamese/English support and frontend Text-to-Speech (TTS) playback.
- Developed an admin dashboard for managing menus, orders, staff, revenue, while applying SEO optimization to customer-facing restaurant pages.
- [Github Backend](#) – [GitHub Frontend](#) – [Live demo](#)

ShortLink Tool – Scalable URL Shortener System

Oct 2025 – Dec 2025

| React.js, Nest.js, MongoDB, Redis, Docker, Nginx, k6

3 months

- Designed and implemented a scalable URL shortening architecture using Base62 and a Key Generation Service (KGS) to pre-generate unique short codes and minimize collision risk.
- Optimized read-heavy redirect flows with Redis caching, null caching, indexed lookups, HTTP 302 redirects, and Redis-backed rate limiting to reduce database load and protect APIs.
- Built an asynchronous analytics pipeline using Redis counters and scheduled batch persistence, keeping analytics processing outside the redirect-critical path
- Evaluated production redirect performance with k6 under staged workloads of up to **100 concurrent VUs**; a mixed hot/cold traffic scenario processed **54K+ redirects with 100% successful HTTP 302 responses, 127 ms p95 and 206 ms p99 latency**.
- Containerized the application with Docker and implemented CI/CD with GitHub Actions, configuring Nginx for reverse proxying, request routing, and production deployment on VPS.
- [Github Backend](#) – [Github Frontend](#) – [Live demo](#)

TECHNICAL SKILLS

Languages | JavaScript, TypeScript, SQL, Python

Frontend & Mobile | React.js, Next.js, React Native, Expo, Tailwind CSS

Backend | Node.js, Express.js, Nest.js, RESTful APIs

Databases | PostgreSQL, MySQL, MongoDB

Tools & DevOps | Git, Docker, Nginx, Postman, Swagger, AWS EC2/S3, Figma, Render, Vercel

AI-assisted Development | ChatGPT, Codex, Antigravity with output verification through testing and documentation.

CERTIFICATION & AWARDS

Academic Encouragement Scholarship DUT

Semester 2, 2024–2025

Semester GPA 9.26/10

TOEIC Listening & Reading

Jun 2026

Score 815/990

Google for Education - Gemini Certified Student

Feb 2026